

# Parameter efficient handling of acquisition artefacts to map tropical forests

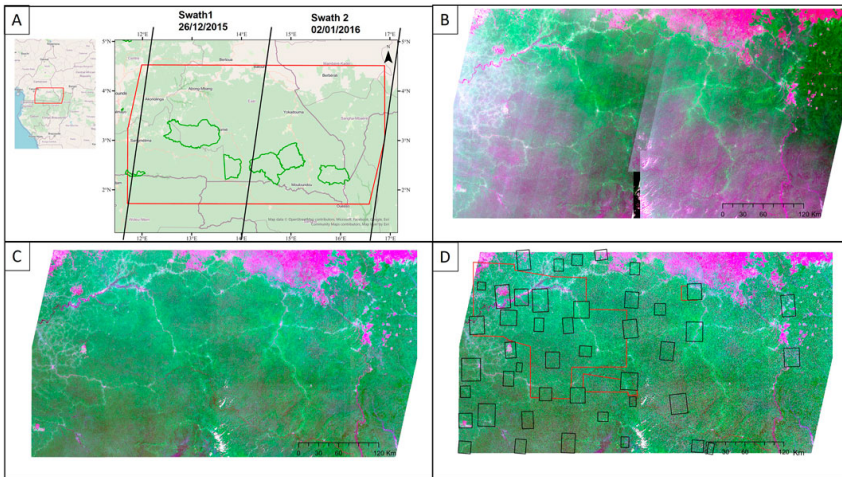
Biomonweek 2026

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Paul Tresson et al.

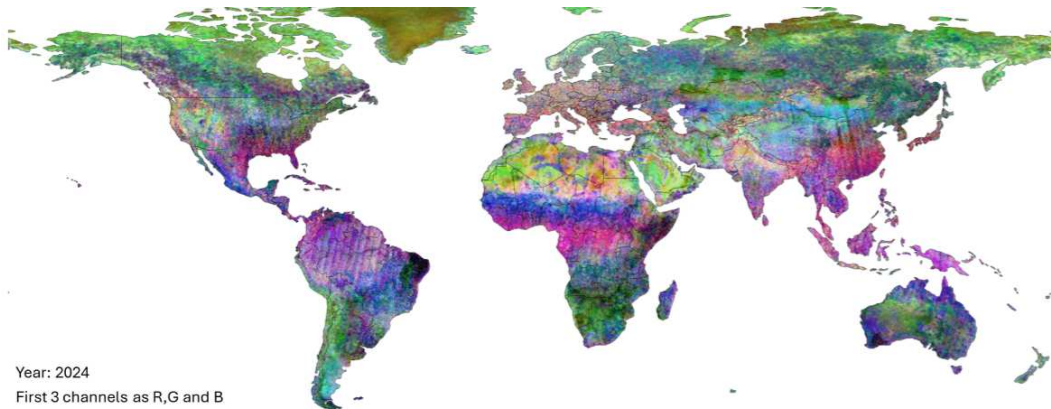
Institut de Recherche pour le Développement - UMR AMAP

# Context: Mapping tropical forests is hard



Viennois et al., 2025

## Context: Omnipotent foundation models ?



From Feng et al., 2025

## Context: Omnipotent foundation models ?

- Not always at local scale
- Not always on niche tasks
- Not for all modalities
- Not always open source and open weights



Can we handle acquisition artefacts to map tropical forests ?

...

Can we handle acquisition artefacts to map tropical forests ?

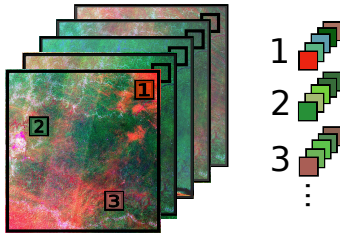
Without training a new expensive model

# Constraints

- Robust
- Versatile
- Actionnable without HPC
- Actionnable with limited labelled dataset

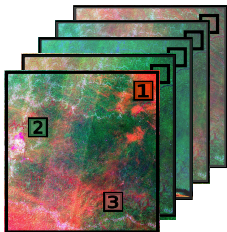
# Methods : overview

## Sampling

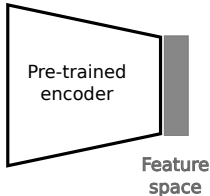


# Methods : overview

## Sampling



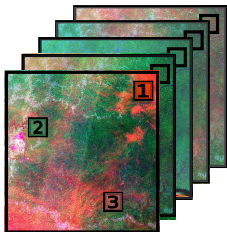
## Model



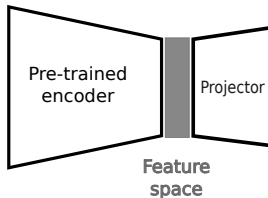
Encoders : DINOv3 and DOFA Siméoni et al., 2025; Xiong et al., 2024

# Methods : overview

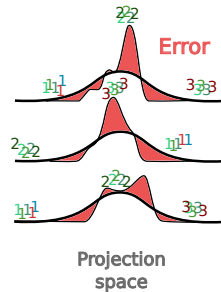
## Sampling



## Model



## Task

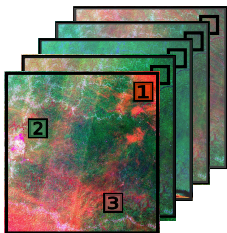


Based on LeJEPA Balestrierio and LeCun, 2025

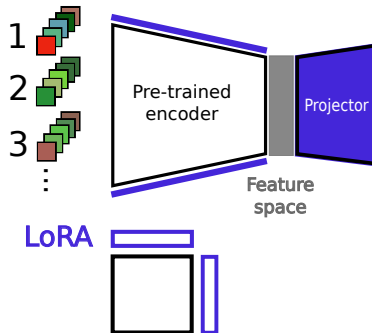


# Methods : overview

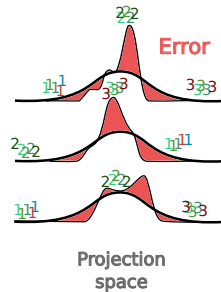
## Sampling



## Model



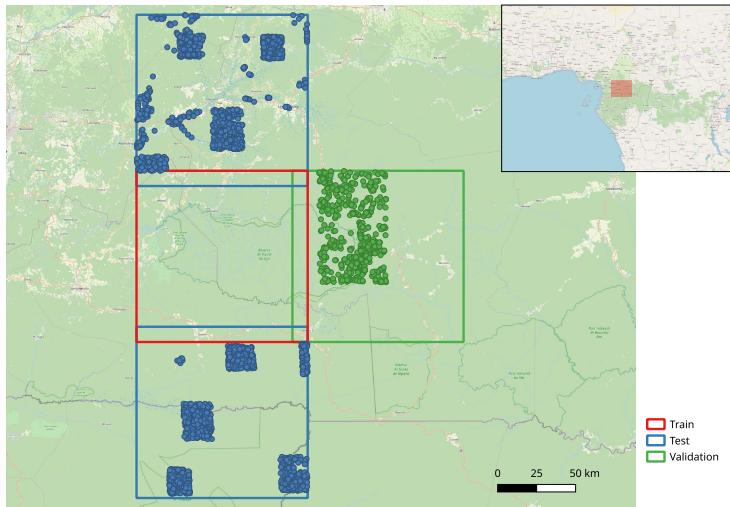
## Task



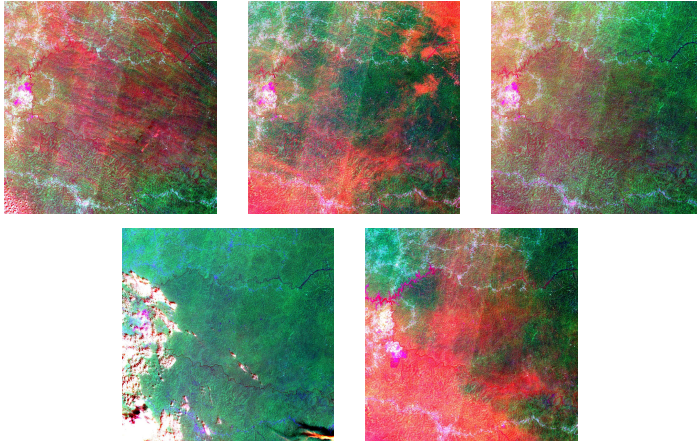
LoRA see Hu et al., 2022

# Methods : datasets

- S2 tiles
- 655 labeled validation points
- 2103 labeled test points
- less than 15% clouds



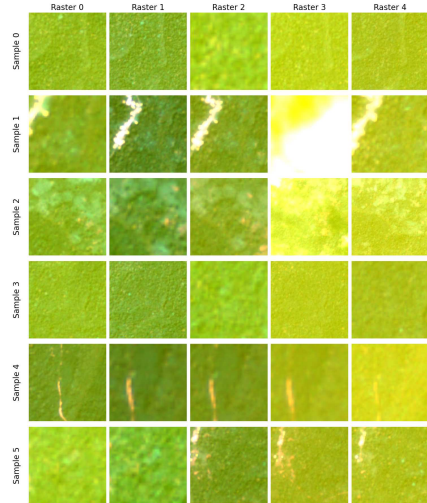
## Methods : train dataset



RGB composition with bands 2, 8 and 11

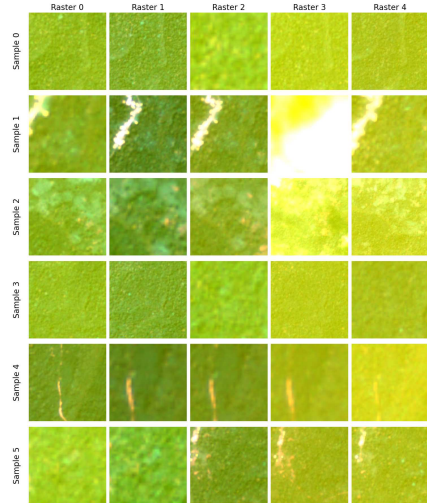
# Methods : train sampling

- Random sampling on 5 dates



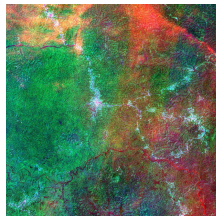
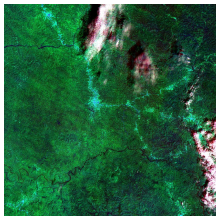
# Methods : train sampling

- Random sampling on 5 dates
- Random zoom and crop  
(Between 30-150 pixels)

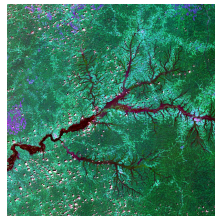
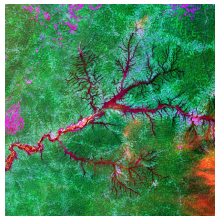
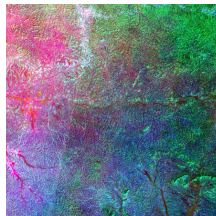
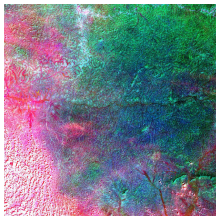


# Methods : validation and test datasets

Validation :

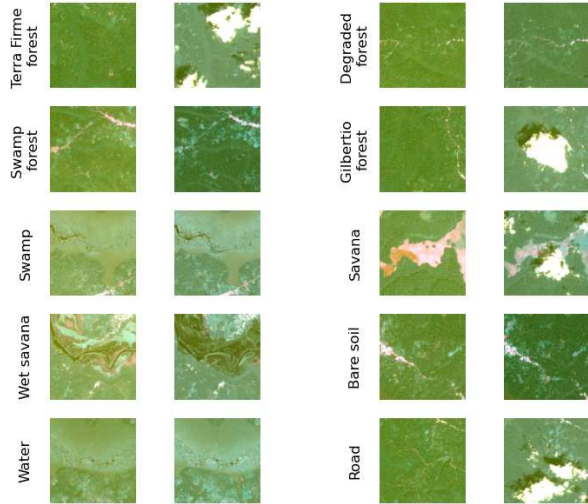


Test :



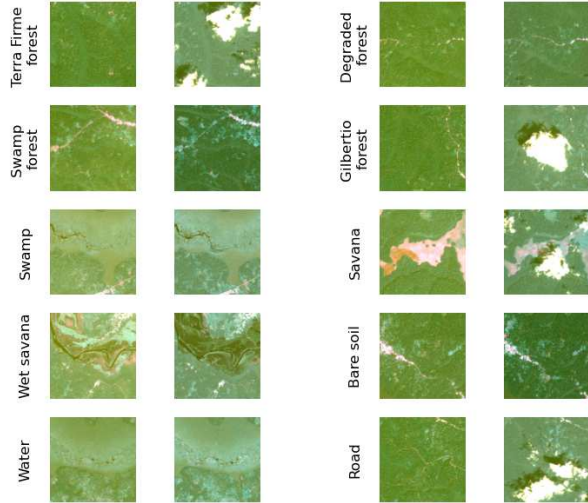
# Methods : validation scheme

- 150 px centered on labeled point



# Methods : validation scheme

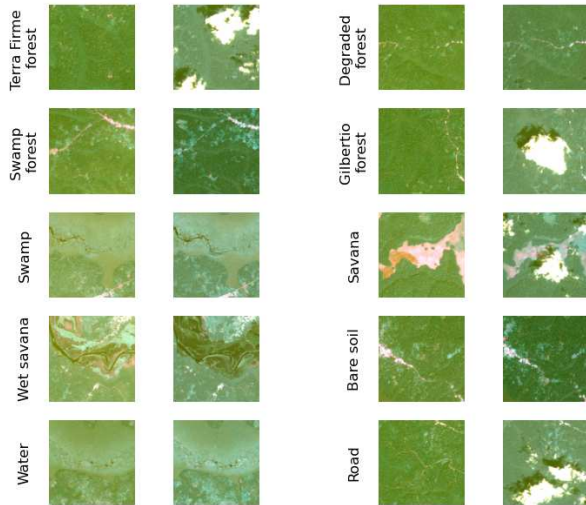
- 150 px centered on labeled point
- Geographical stratified 5-fold





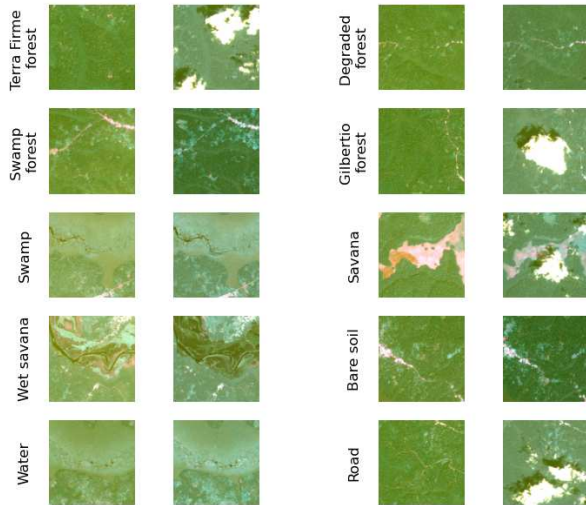
# Methods : validation scheme

- 150 px centered on labeled point
- Geographical stratified 5-fold
- KNN fitted on model embedding of version A ...

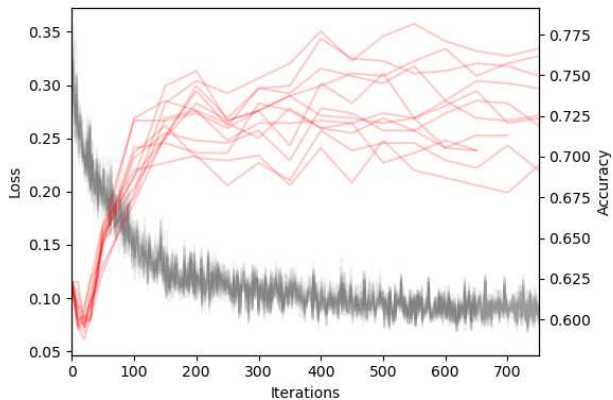


# Methods : validation scheme

- 150 px centered on labeled point
- Geographical stratified 5-fold
- KNN fitted on model embedding of version A ...  
... and tested on version B



## Results : training behaviour



# Results : Confusion matrix

Before training

| True label \ Predicted label | Build | Degraded Forest | Gilbertio. Forest | Road | Savana | Swamp | Swamp forest | Terra Firme Forest | Water | Wet savana |
|------------------------------|-------|-----------------|-------------------|------|--------|-------|--------------|--------------------|-------|------------|
| Build                        | 24    | 77              | 1                 | 4    | 2      | 1     | 26           | 2                  | 0     | 0          |
| Degraded Forest              | 28    | 112             | 37                | 2    | 3      | 0     | 99           | 43                 | 2     | 0          |
| Gilbertio. Forest            | 2     | 7               | 84                | 0    | 0      | 0     | 150          | 43                 | 0     | 0          |
| Road                         | 8     | 17              | 3                 | 1    | 4      | 0     | 9            | 2                  | 0     | 0          |
| Savana                       | 2     | 9               | 0                 | 2    | 79     | 1     | 6            | 5                  | 2     | 8          |
| Swamp                        | 2     | 2               | 0                 | 0    | 7      | 2     | 0            | 0                  | 5     | 11         |
| Swamp forest                 | 34    | 160             | 193               | 3    | 12     | 0     | 141          | 155                | 7     | 0          |
| Terra Firme Forest           | 9     | 36              | 91                | 1    | 3      | 0     | 169          | 105                | 1     | 0          |
| Water                        | 0     | 5               | 0                 | 0    | 7      | 1     | 9            | 0                  | 3     | 2          |
| Wet savana                   | 0     | 0               | 0                 | 0    | 5      | 6     | 0            | 0                  | 1     | 8          |

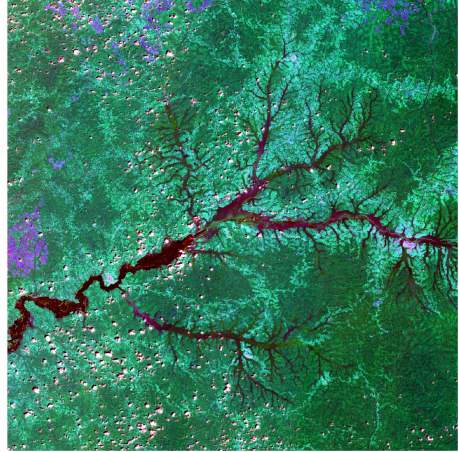
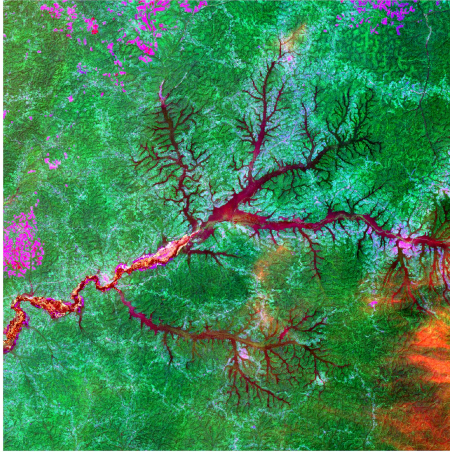
After training

| True label \ Predicted label | Build | Degraded Forest | Gilbertio. Forest | Road | Savana | Swamp | Swamp forest | Terra Firme Forest | Water | Wet savana |
|------------------------------|-------|-----------------|-------------------|------|--------|-------|--------------|--------------------|-------|------------|
| Build                        | 79    | 41              | 0                 | 8    | 2      | 0     | 6            | 1                  | 0     | 0          |
| Degraded Forest              | 39    | 152             | 10                | 4    | 1      | 0     | 52           | 68                 | 0     | 0          |
| Gilbertio. Forest            | 0     | 3               | 133               | 0    | 0      | 0     | 105          | 45                 | 0     | 0          |
| Road                         | 26    | 10              | 0                 | 2    | 0      | 0     | 2            | 4                  | 0     | 0          |
| Savana                       | 1     | 1               | 0                 | 3    | 93     | 1     | 1            | 2                  | 0     | 12         |
| Swamp                        | 3     | 0               | 0                 | 0    | 9      | 8     | 0            | 0                  | 2     | 7          |
| Swamp forest                 | 3     | 22              | 102               | 1    | 0      | 0     | 521          | 38                 | 18    | 0          |
| Terra Firme Forest           | 2     | 24              | 41                | 0    | 0      | 0     | 36           | 312                | 0     | 0          |
| Water                        | 0     | 0               | 0                 | 0    | 0      | 5     | 4            | 0                  | 13    | 5          |
| Wet savana                   | 0     | 0               | 0                 | 0    | 8      | 4     | 0            | 0                  | 2     | 6          |

36% accuracy gain !

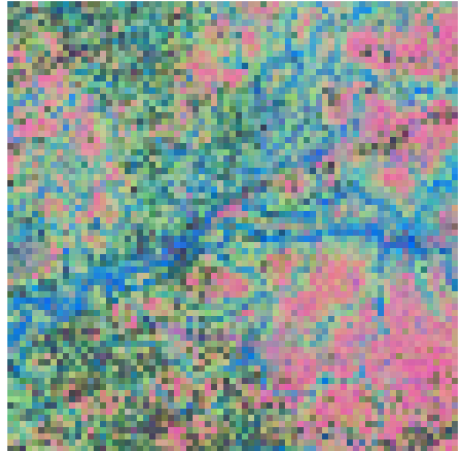
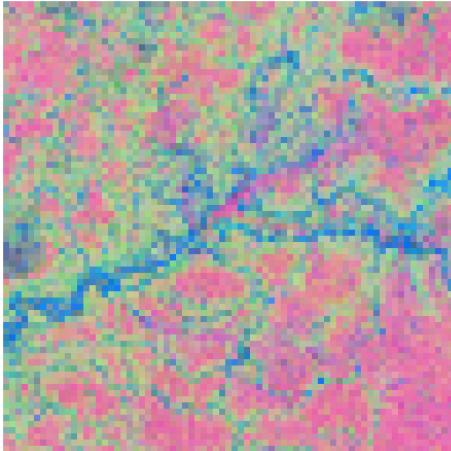
# Visualisation

## Original rasters



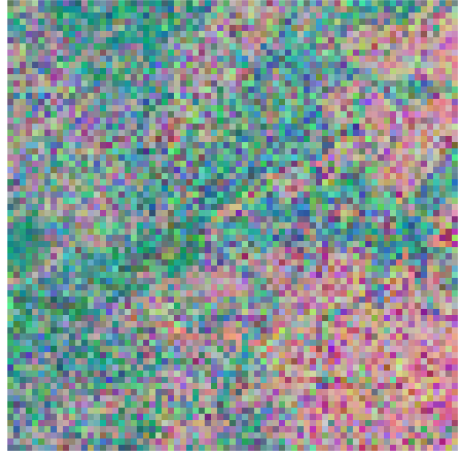
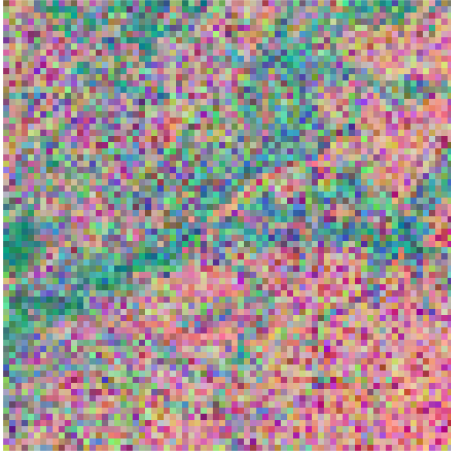
# Visualisation : PCA

Before training



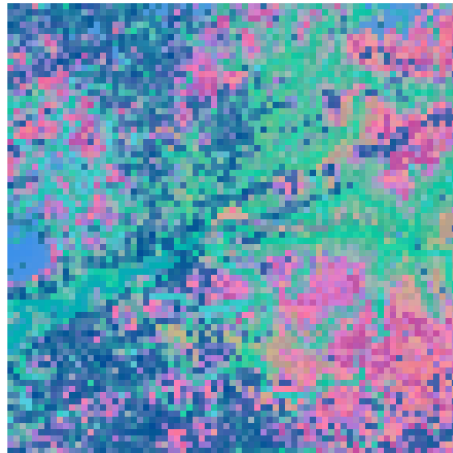
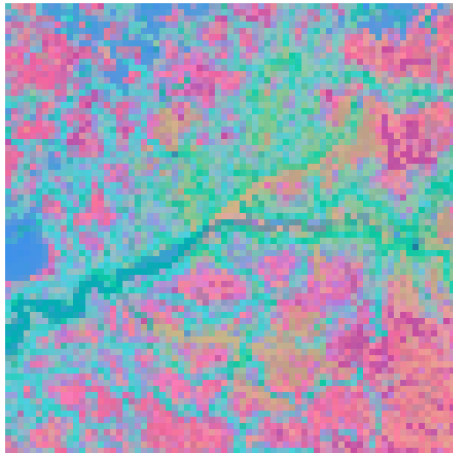
## Visualisation : PCA

After training



# Visualisation : UMAP

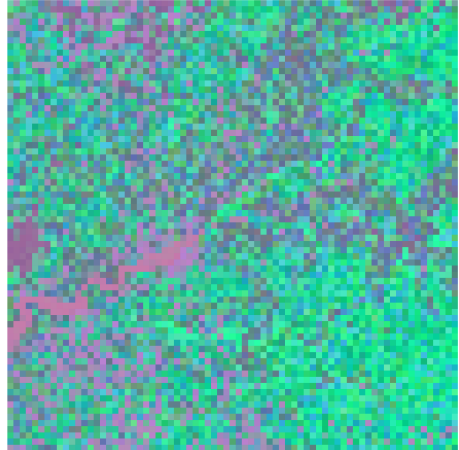
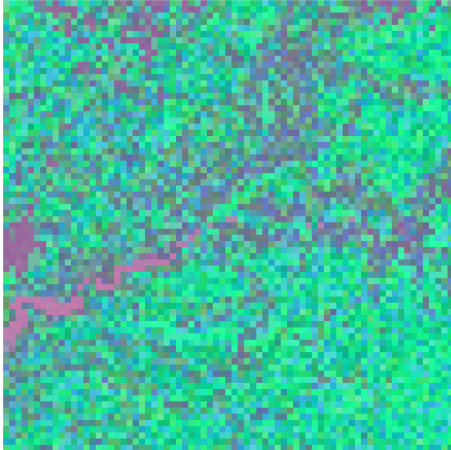
Before training





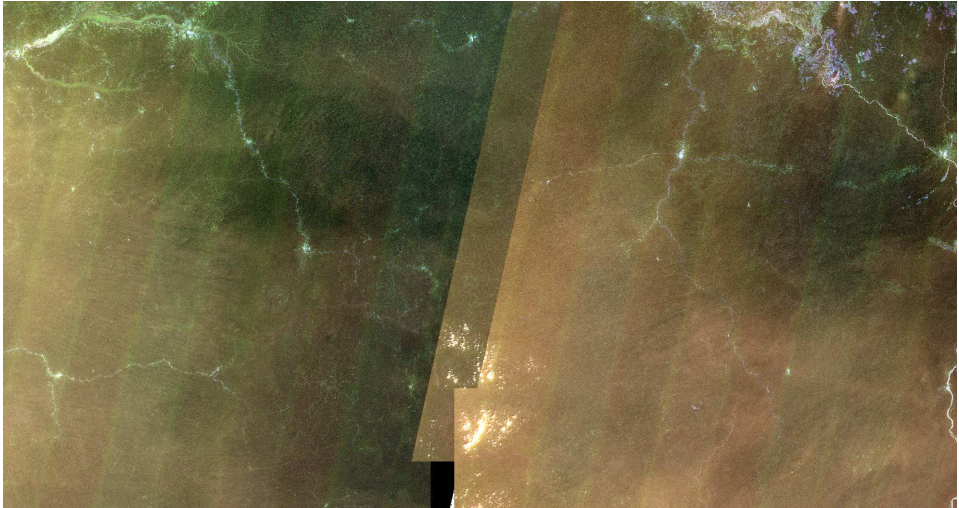
# Visualisation : UMAP

After training



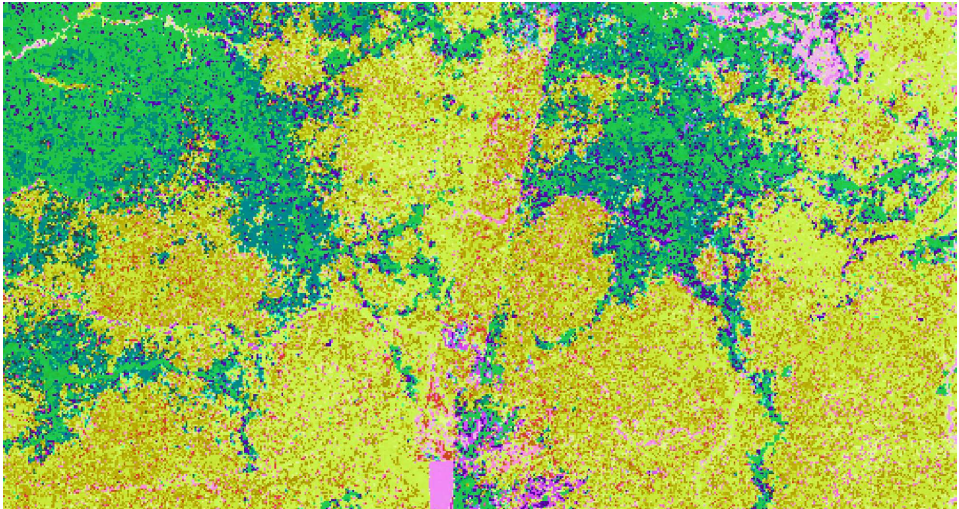
# Scaling up

Original



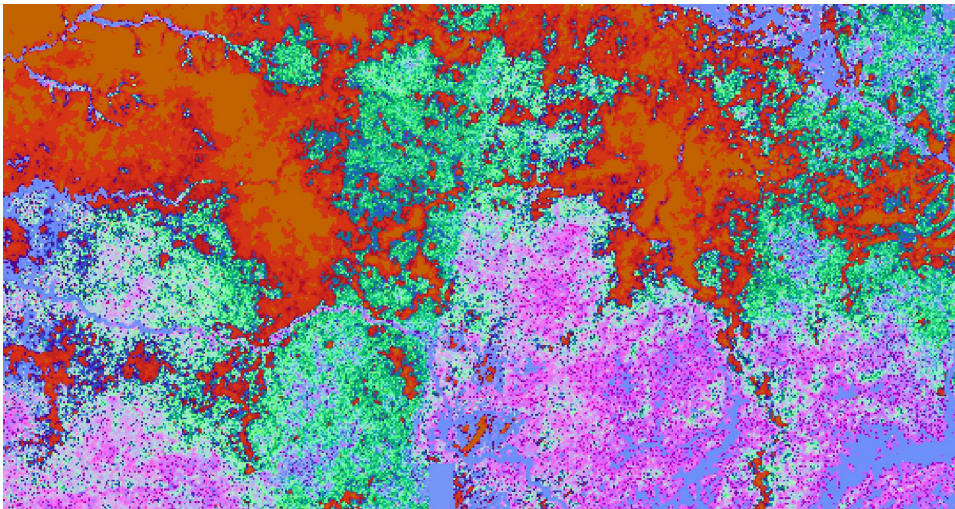
# Scaling up

Before training

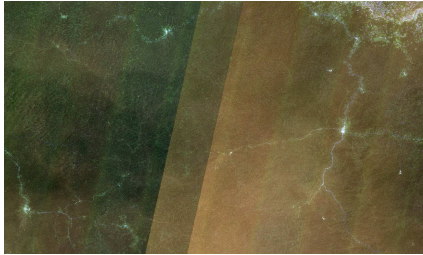


## Scaling up

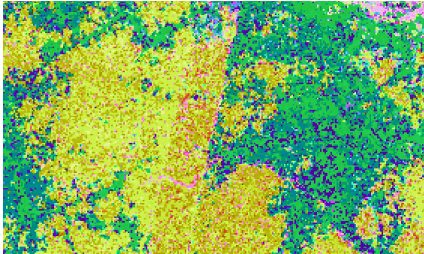
After training



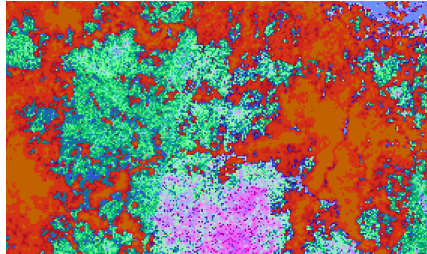
## Scaling up



Before



After



# Discussion

Not perfect yet but

- Only a couple hundred iterations (not epochs)

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0.9 / 85.6M



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## Not perfect yet but

- Only a couple hundred iterations (not epochs)
- No labels needed
- Training only  $\approx 1\%$  of the weights on a single GPU  
0.9 / 85.6M
- Valid for any raster data

# Discussion

## Not perfect yet but

- Only a couple hundred iterations (not epochs)
- No labels needed
- Training only  $\approx 1\%$  of the weights on a single GPU  
0.9 / 85.6M
- Valid for any raster data
- To map something **that does not vary between versions**

# Thanks for your attention !

## Any questions ?

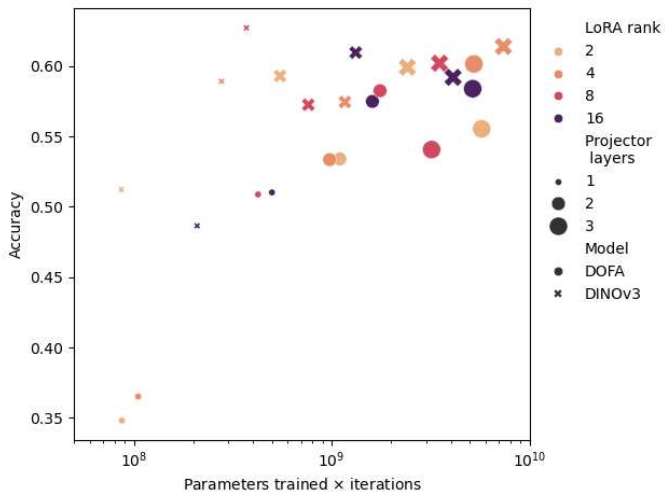
## References i

- Balestriero, Randall and Yann LeCun (2025). **“Lejepa: Provable and scalable self-supervised learning without the heuristics”**. In: *arXiv preprint arXiv:2511.08544*.
- Feng, Zhengpeng et al. (2025). **“Tessera: Temporal embeddings of surface spectra for earth representation and analysis”**. In: *arXiv preprint arXiv:2506.20380*.
- Hu, Edward J et al. (2022). **“Lora: Low-rank adaptation of large language models.”**. In: *ICLR 1.2*, p. 3.
- Siméoni, Oriane et al. (2025). **“Dinov3”**. In: *arXiv preprint arXiv:2508.10104*.
- Viennois, Gaëlle et al. (2025). **“Sentinel-2 forest typology mapping in Central Africa: assessing deep learning and image preprocessing effects”**. In: *Frontiers in Remote Sensing 6*, p. 1682132.

## References ii

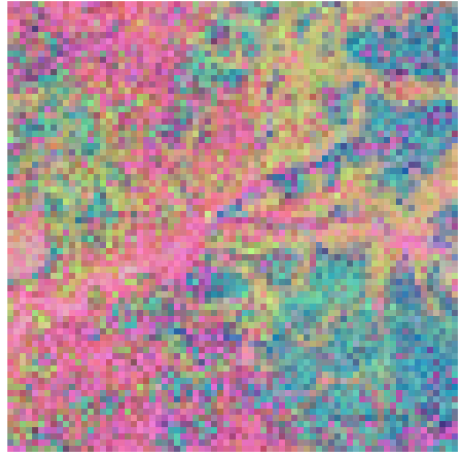
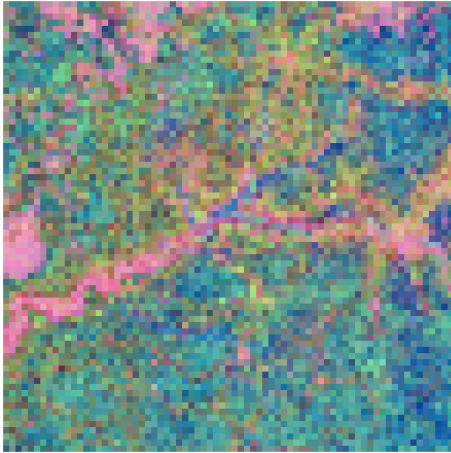
Xiong, Zhitong et al. (2024). “Neural Plasticity-Inspired Foundation Model for Observing the Earth Crossing Modalities”. In: *arXiv preprint arXiv:2403.15356*.

## Results : Accuracy vs trained parameters



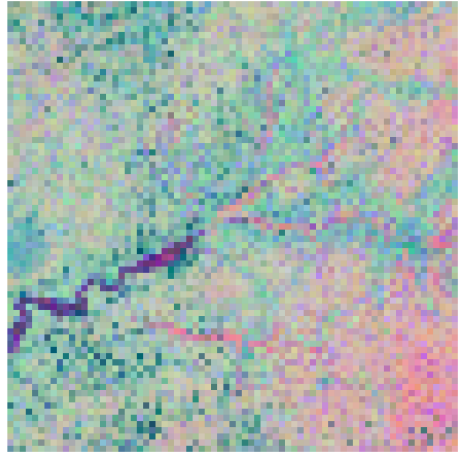
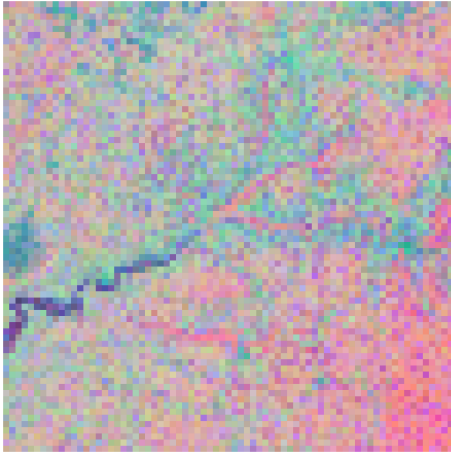
# Visualisation DOFA PCA

Before training



# Visualisation DOFA PCA

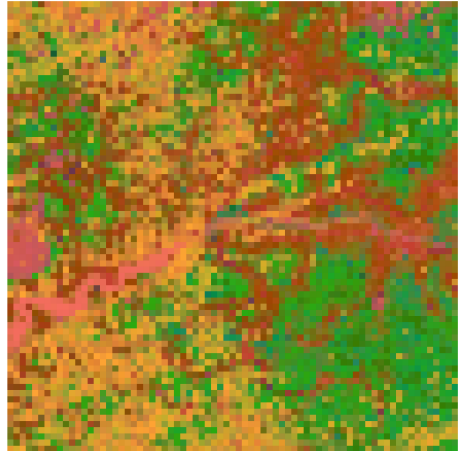
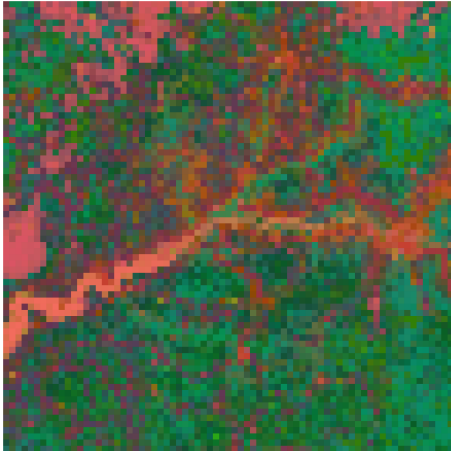
After training





# Visualisation DOFA UMAP

Before training



# Visualisation DOFA UMAP

After training

